TRANSICIONES DE FASE EN EL MODELO DE HEISENBERG APLICADO A BICAPAS

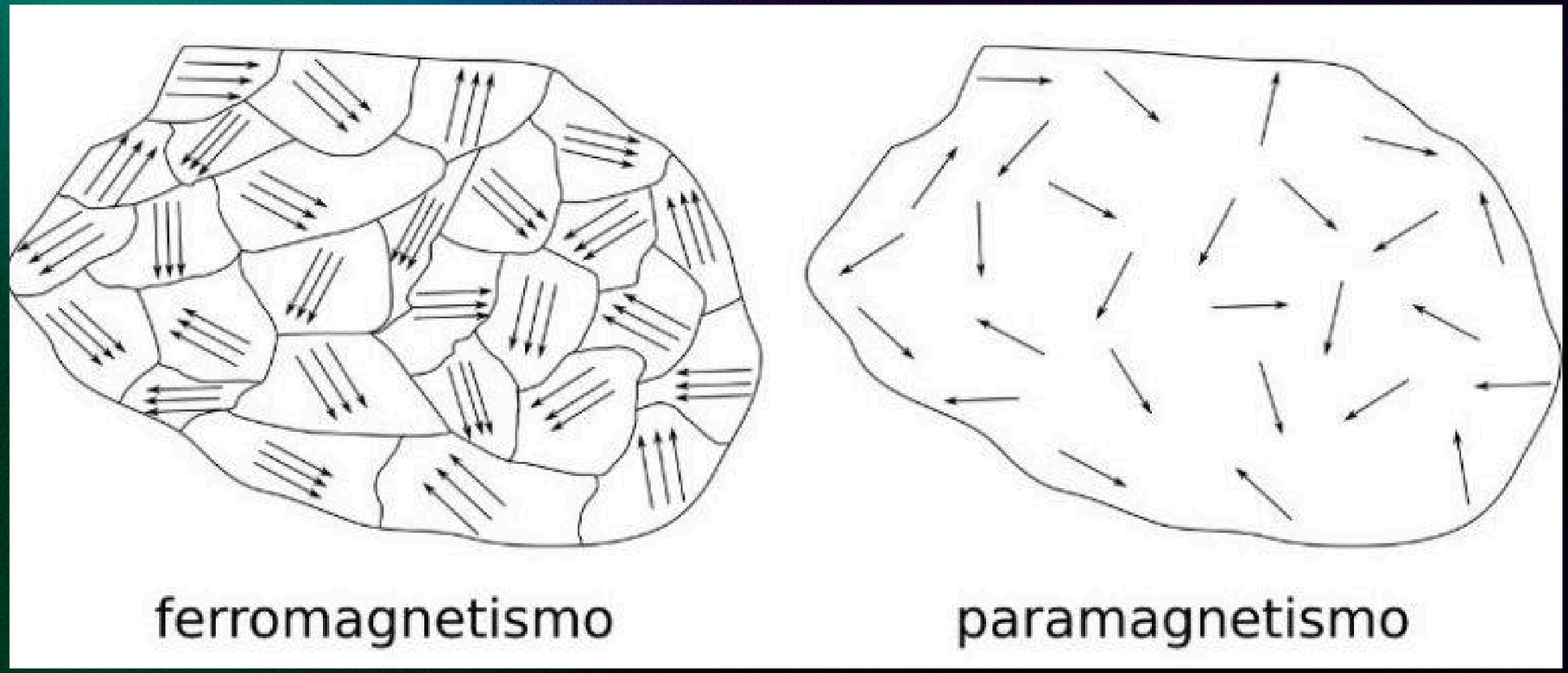
Jose Daniel Villada Godoy †

Física computacional i 2025-2

TEMAS

1. INTRODUCCIÓN
2. MUESTERO DE MORSAGLIA
3. M VS T: BICAPA FERRO/ANTIFERRO
4. M VS T: BICAPA FERRO/PARAMAGNETICA
5. CONCLUSIONES

INTRODUCCION



INTRODUCCIÓN

OBJETIVO: DETERMINAR EL COMPORTAMIENTO DE LOS SISTEMAS (1) FERRO/ANTIFERRO Y (2) FERRO/PARAMAGNETICO USANDO ALGORITMO DE METRÓPOLIS.

$$P_{\text{aceptar}} = \min\left(1,\ e^{-\Delta E/T}\right)$$

$$\mathcal{H} = -J_F \sum_{\langle i,j\rangle \in F} \mathbf{S}_i^{(1)} \cdot \mathbf{S}_j^{(1)} - J_I \sum_i \mathbf{S}_i^{(1)} \cdot \mathbf{S}_i^{(2)} - \sum_{\alpha=1}^{2} \sum_i \mathbf{H} \cdot \mathbf{S}_i^{(\alpha)}$$

$$\mathcal{H} = -J_F \sum_{\langle i,j\rangle \in F} \mathbf{S}_i^{(1)} \cdot \mathbf{S}_j^{(1)} + J_{AF} \sum_{\langle i,j\rangle \in AF} \mathbf{S}_i^{(2)} \cdot \mathbf{S}_j^{(2)} - J_I \sum_i \mathbf{S}_i^{(1)} \cdot \mathbf{S}_i^{(2)} - \sum_{\alpha=1}^{2} \sum_i \mathbf{H} \cdot \mathbf{S}_i^{(\alpha)}$$

MUESTREAR LA ESFERA: ALGORITMO DE MORSAGLIA

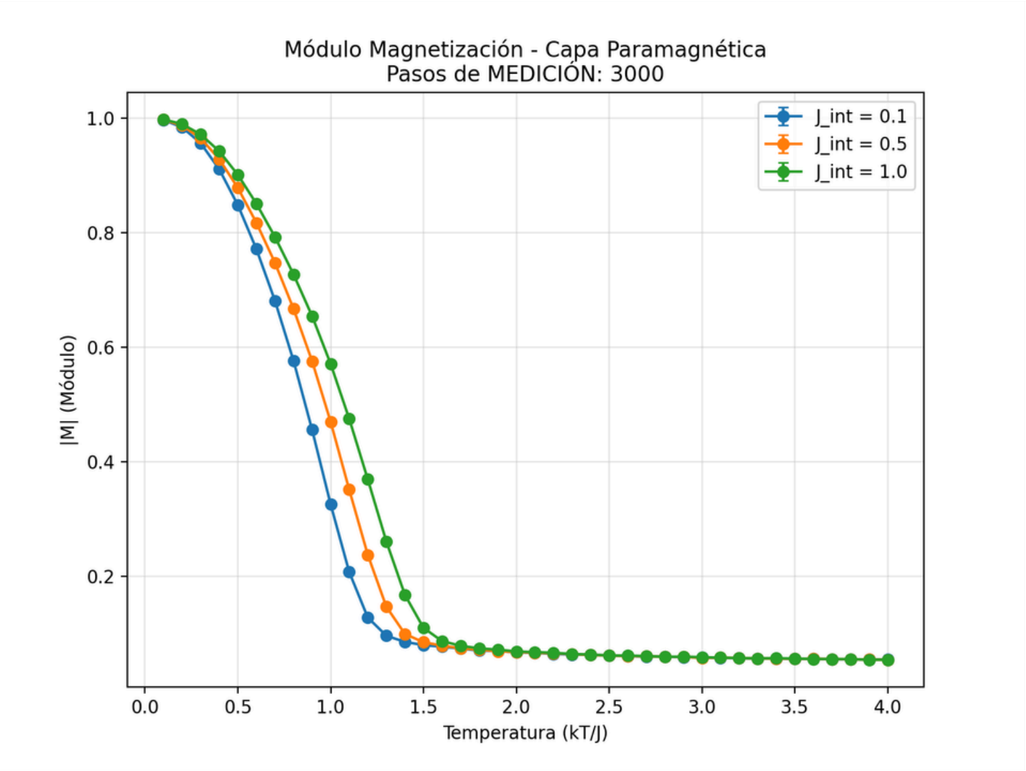
$$S_x = 2\zeta_1(1-\zeta^2)^{1/2}, \qquad S_y = 2\zeta_2(1-\zeta^2)^{1/2}, \qquad S_z = 1-2\zeta^2.$$

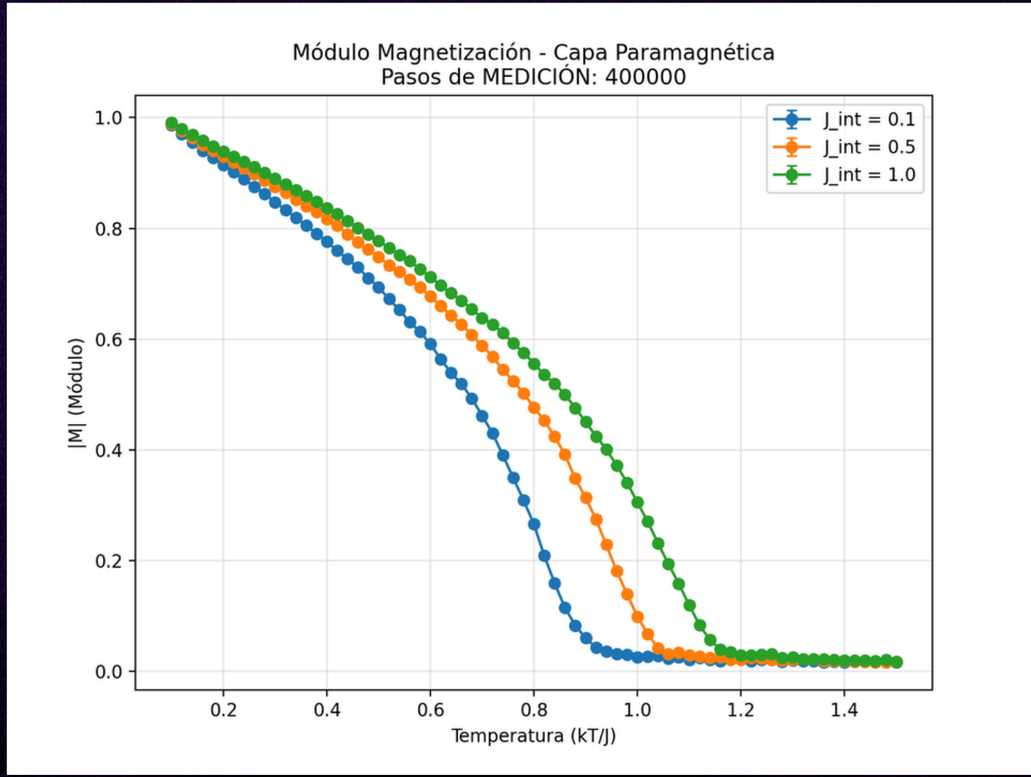
$$\zeta_1 = 1-2r_1 \text{ and } \zeta_2 = 1-2r_2$$

LOS VALORES SE ACEPTAN SI

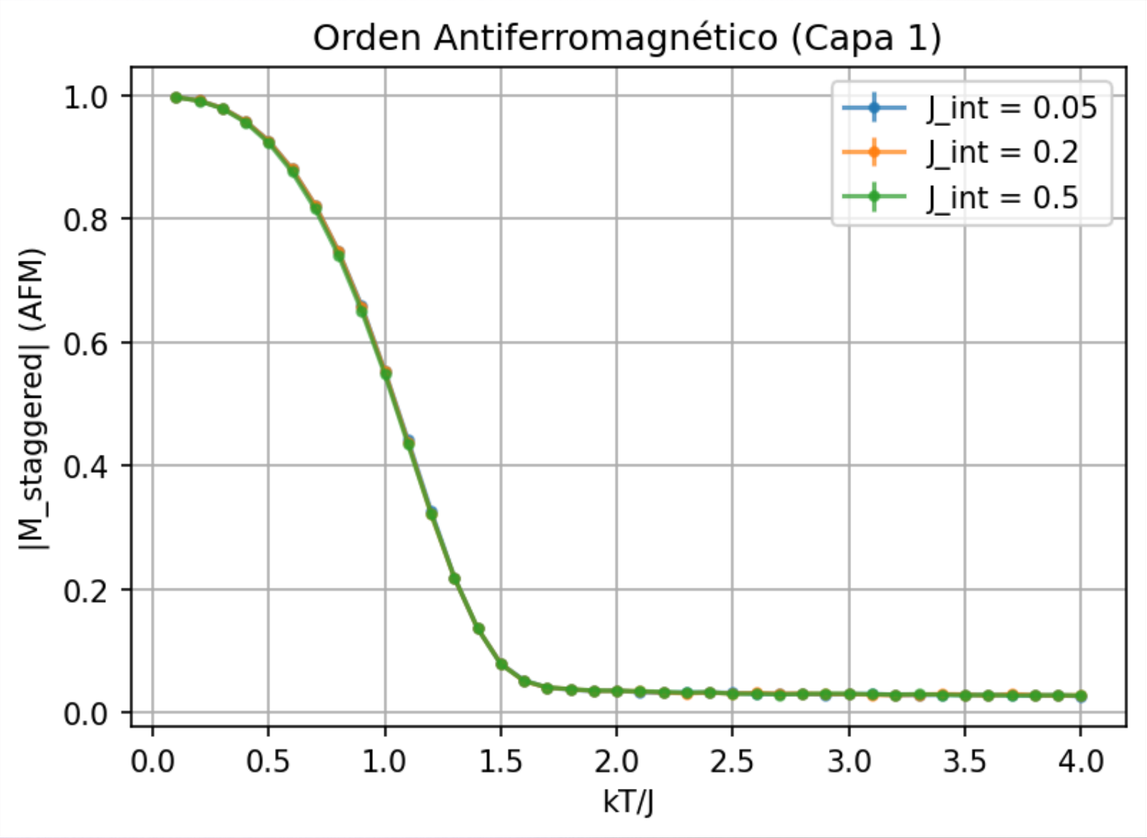
$$\zeta^2 = \zeta_1^2 + \zeta_2^2 \text{ and if } \zeta^2 < 1$$

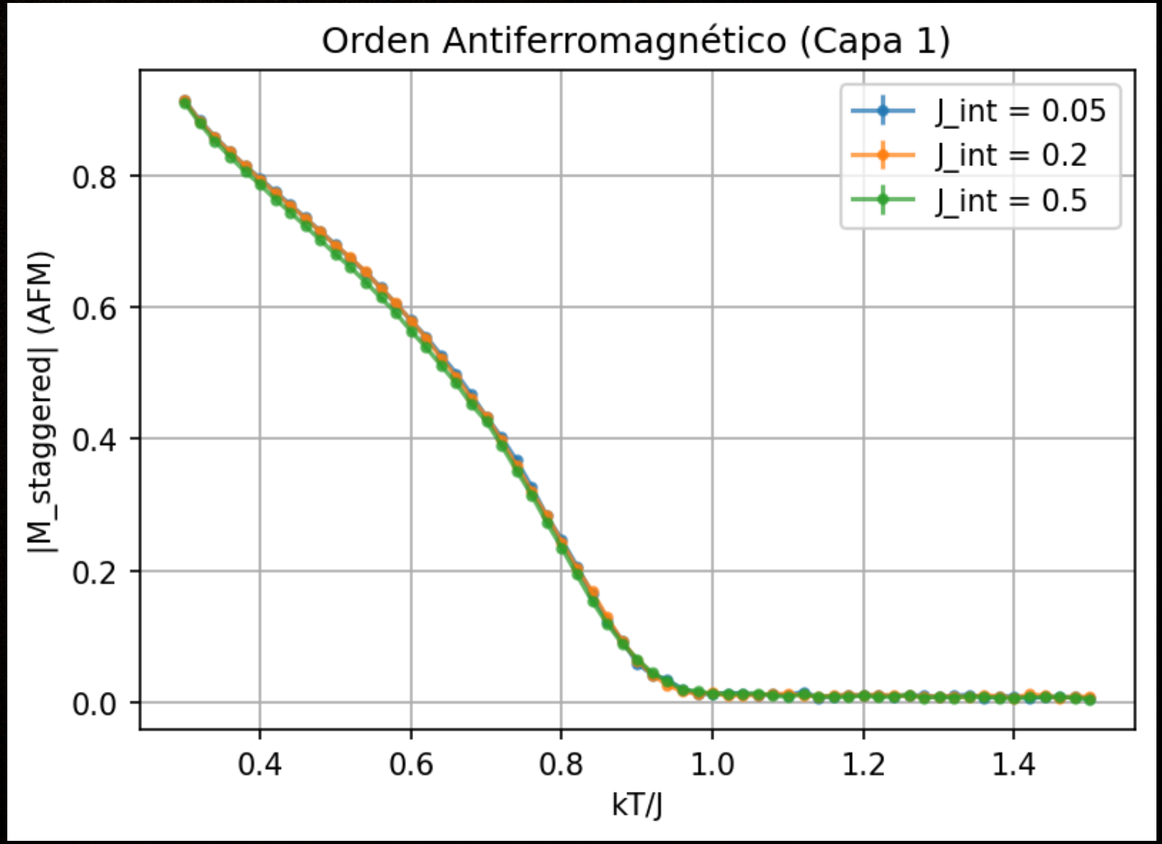
BICAPA PARAMAGNÉTICA



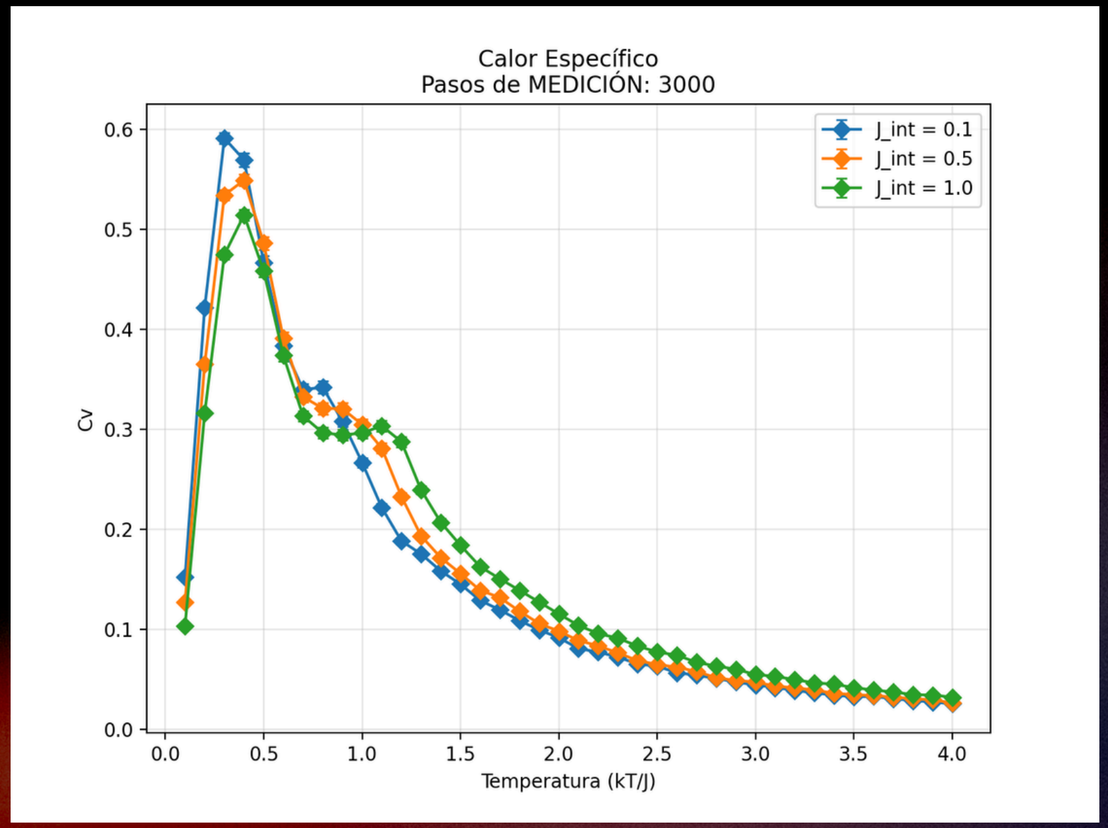


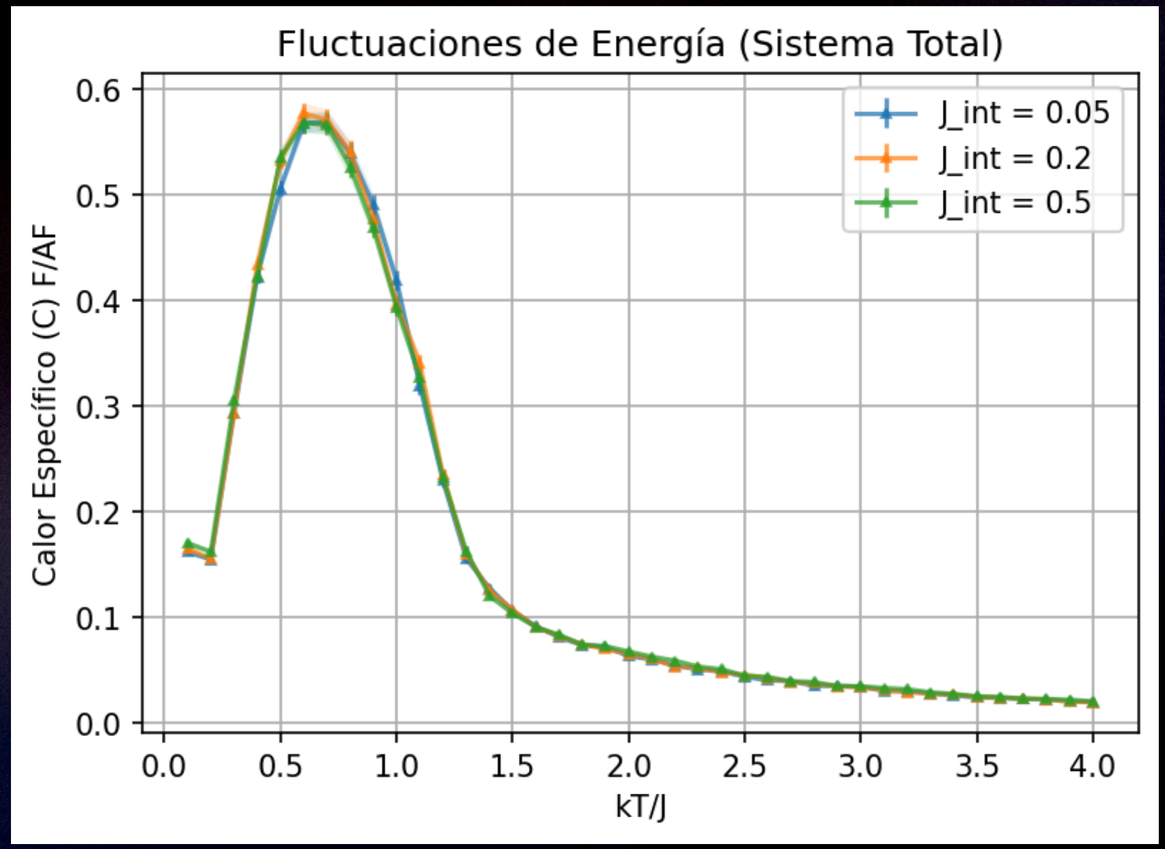
BICAPA FERRO/ANTIFERRO



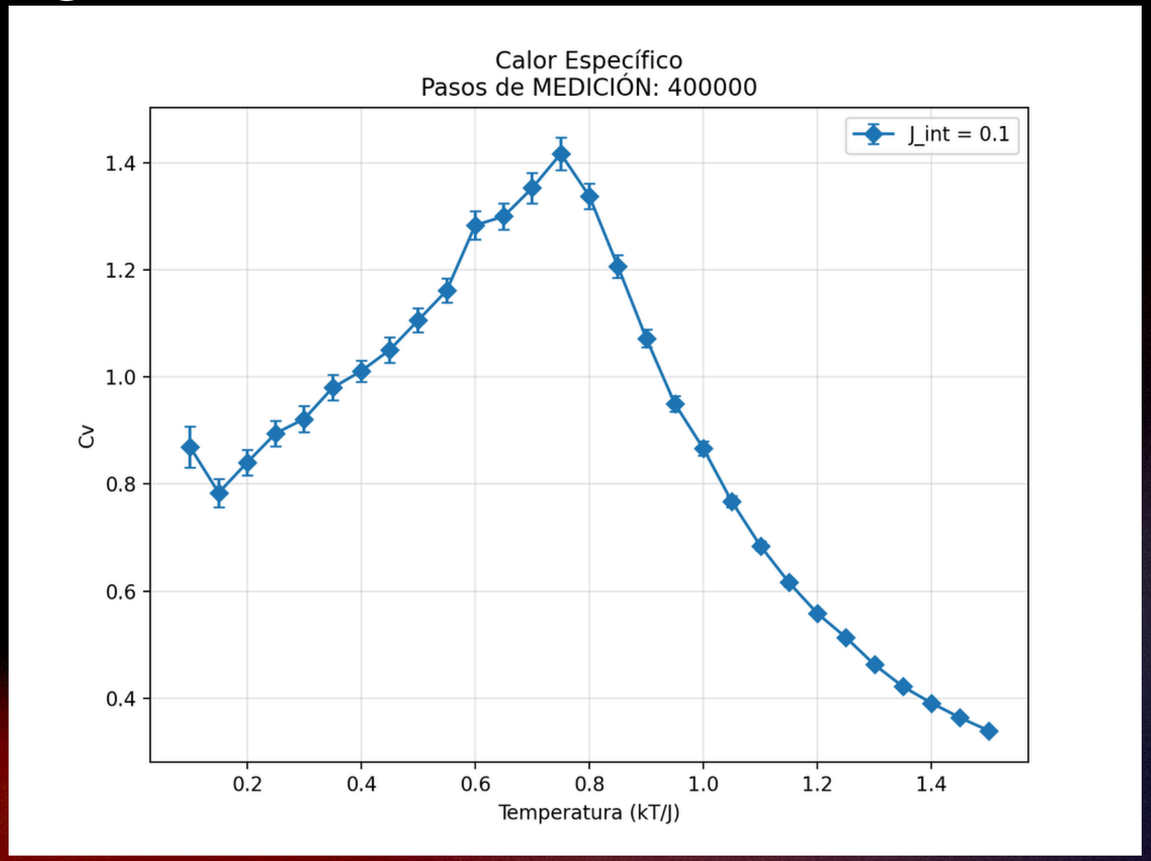


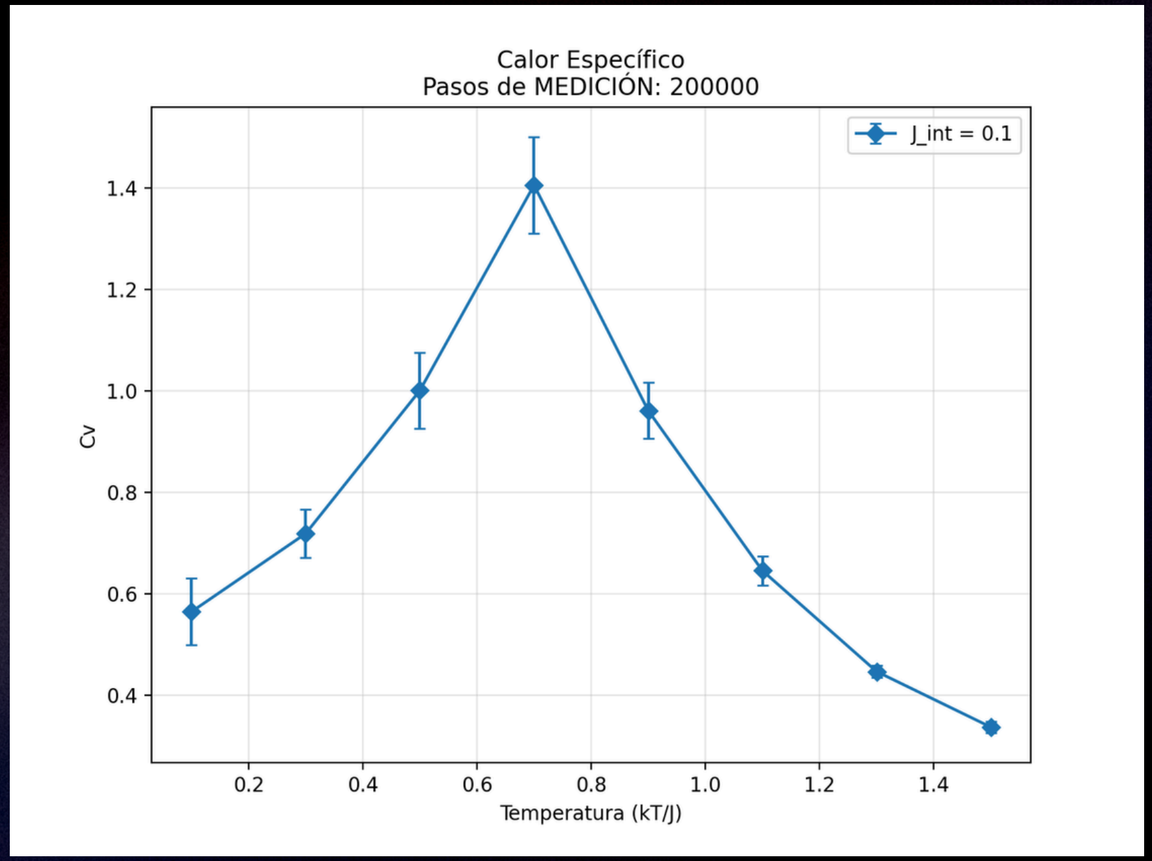
EN BÚSQUEDA DE LA TRANSICION DE FASE: LA CAPACIDAD CALORÍFICA. NO HAY TRANSICIÓN DE FASE POR EL TEOREMA DE MERMIN WAGNER [4]





SOLO HAY TRANSICIÓN DE FASE SI HAY TÉRMINO ZEEMAN (¿?)





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- [3] J. RESTREPO, E. V. ALBANO Y K. BINDER, MAGNETIC PROPERTIES OF FERRO–ANTIFERROMAGNETIC BILAYERS, PHYS. REV. B 85, 054432 (2012).
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- [5] CÓDIGO: HTTPS://GITHUB.COM/ELGOD17/BICAPAS-HEISENBERG-METROPOLIS